

Detecting Synthetic Donald Trump Tweets (Real vs. Generated Text)

Course: Deep Learning (Spring Semester 2026)

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Deep Learning SS26

ACLS

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Project Objective

The goal of this project is to generate authentic-sounding tweets in the style of Donald Trump.

Data Sourcing & preprocessing

Source : The "Donald Trump Tweets Dataset" on ([Kaggle](#)).

Initial Dataset: **56,571** raw records.

Filtering: Removed retweets, deleted isolated URLs, cleaned encodings.

Result: **45,454** real tweets



Donald J. Trump ✓
@realDonaldTrump

A whole civilization will die tonight, never to be brought back again. I don't want that to happen, but it probably will. However, now that we have Complete and Total Regime Change, where different, smarter, and less radicalized minds prevail, maybe something revolutionarily wonderful can happen, WHO KNOWS? We will find out tonight, one of the most important moments in the long and complex history of the World. 47 years of extortion, corruption, and death, will finally end. God Bless the Great People of Iran!

Tokenization

"Sleepy Joe Biden" -> ["Sleepy", "Joe", "Biden"]

"America is back!" -> ["America", "is", "back", "!"]

"Massive RED WAVE coming!!!" -> ?

"Many such “deplorable” instances!" -> ?

Statistical Text Generation (N-gram Models)

3-gram	"Lyn' Hillary could be a dummy and Mrs. We are opening what you're overrated, for the last name, we will they renamed it!"
...	...
8-gram	"Biden FAILED BADLY with the Swine Flu Epidemic. He knows nothing. All talk and no action and we are all tired of incompetent politicians and bad deals!"

Higher n-grams are much more coherent.

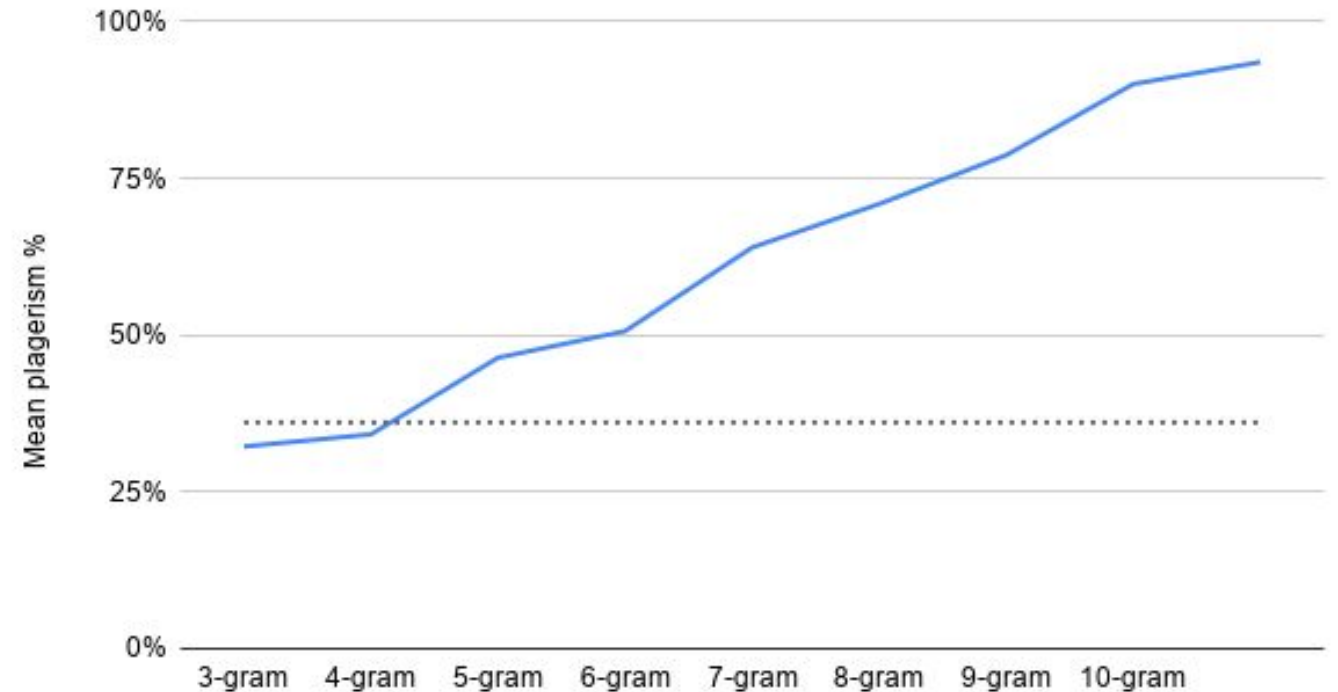
But higher n-grams start to plagiarize real content.

What is the threshold for plagiarization?

Plagiarism Threshold

- Strategy: Calculated the (LCS) to evaluate how much the generator memorizes vs. innovates.
- Baseline: Measured Trump's internal repetition habit across 45k real tweets, establishing a **36%** allowable similarity threshold.
- Selection: the **4-gram** model perfectly hits this optimal boundary maximizing stylistic flavour without crossing into plagiarism.

Plagerism percentage vs n-grams



Project Objective

The Goal of this project is to study how well a deep learning classifier can distinguish real **Donald Trump tweets** from synthetic ones generated by simple language models.

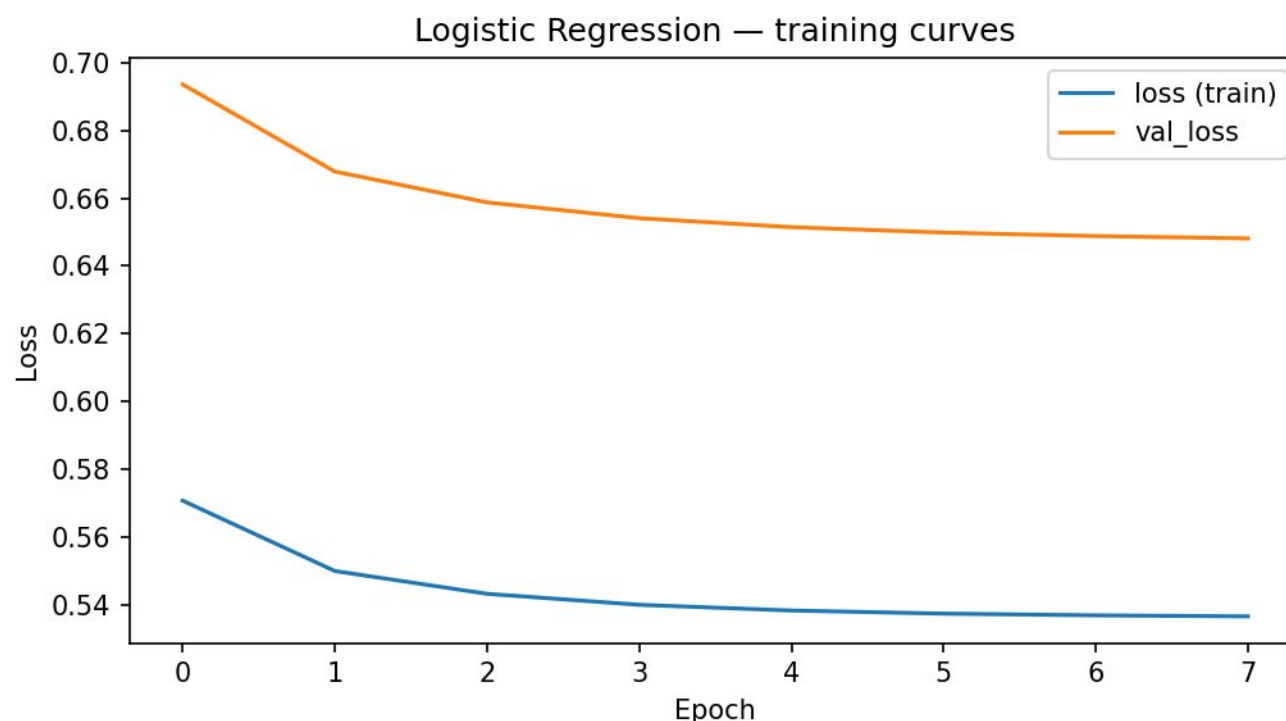
Project Workflow / Methodology

- Phase 1: Statistical Text Generation (N-gram Models)
- Phase 2: Multi-Architecture Classification Training
- Phase 3: Advanced LLM Synthesis
- Phase 4: Ensemble Rejection Sampling and Stress Testing
- Phase 5: Re-classify Re-generated data

- **Phase 2: Multi-Architecture Classification**

To rigorously distinguish real tweets from synthetic ones, we deployed a progression of five distinct architectures, ranging from traditional ML to state-of-the-art Deep Learning:

1. Logistic Regression (TF-IDF):



Classification report				
	precision	recall	f1-score	support
fake	0.58	0.65	0.61	9091
real	0.61	0.53	0.57	9208
accuracy			0.59	18299
macro avg	0.59	0.59	0.59	18299
weighted avg	0.59	0.59	0.59	18299

Running with 8-gram

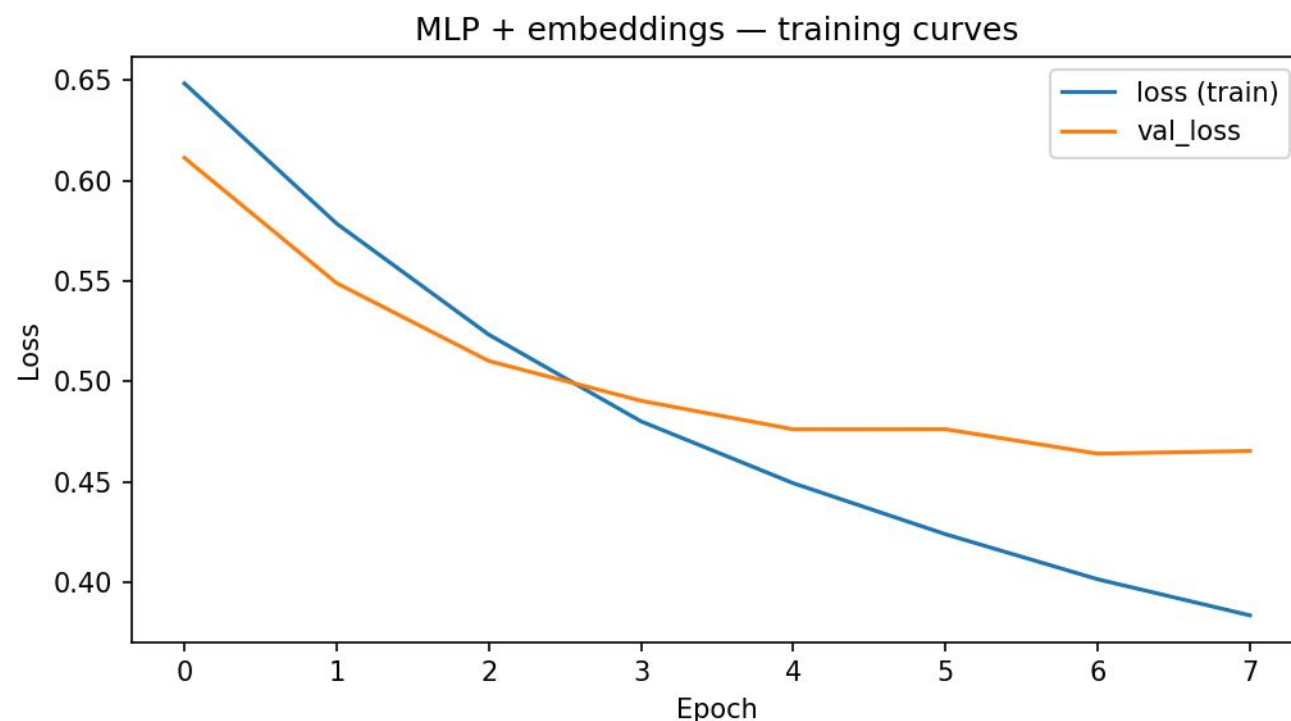
- We developed a fast LCS algorithm.
- Generated 1 million 8-gram tweets.
- Threw away 95% that scored above 36%.
- Got dataset of 45k valid fake tweets.

"Obama's offer to Iran will not stop the very potentially dangerous flights to and from certain parts of the Country that developed vaccines, and years ahead I look forward to attending &, speaking at the @NHGOP #FITN Republican Leadership Summit on Saturday at 12PM! Let's Make America Great Again!"

- **Phase 2: Multi-Architecture Classification**

To rigorously distinguish real tweets from synthetic ones, we deployed a progression of five distinct architectures, ranging from traditional ML to state-of-the-art Deep Learning:

2. Embedding + MLP (Multilayer Perceptron):



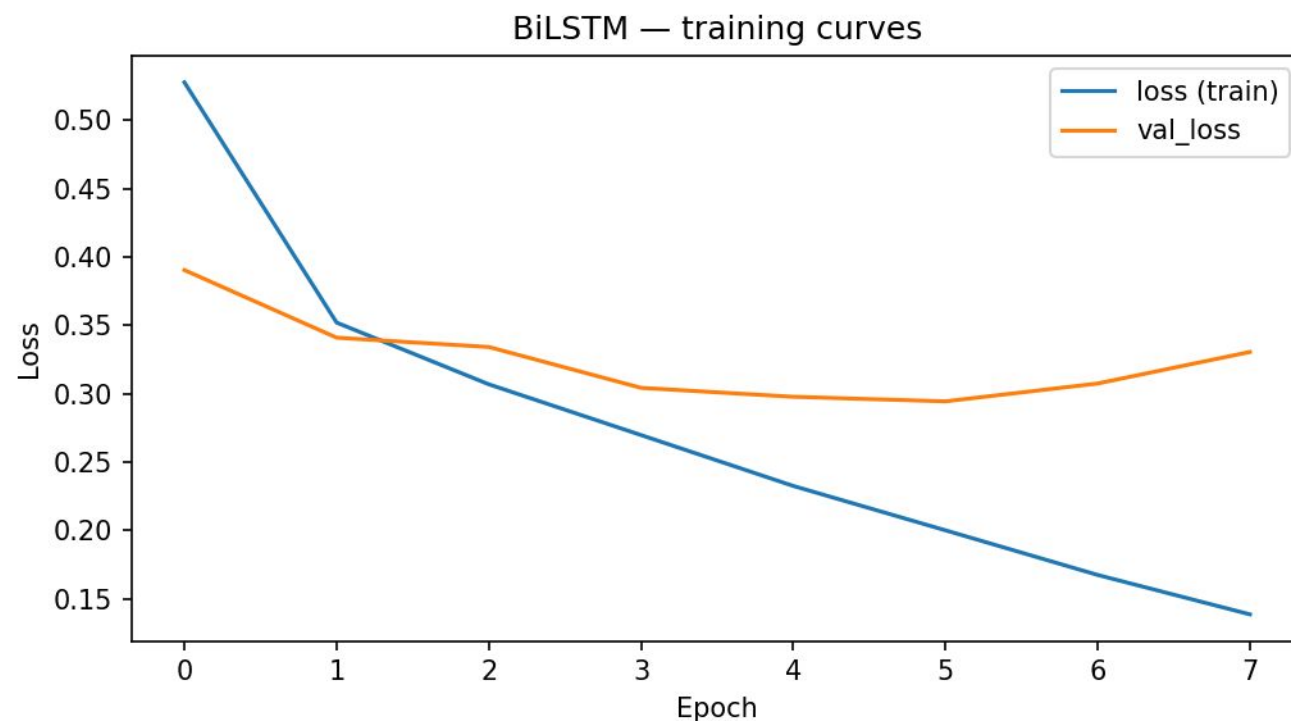
Classification report

	precision	recall	f1-score	support
fake	0.79	0.75	0.77	9091
real	0.77	0.81	0.79	9208
accuracy			0.78	18299
macro avg	0.78	0.78	0.78	18299
weighted avg	0.78	0.78	0.78	18299

- **Phase 2: Multi-Architecture Classification**

To rigorously distinguish real tweets from synthetic ones, we deployed a progression of five distinct architectures, ranging from traditional ML to state-of-the-art Deep Learning:

3. Bidirectional LSTM (BiLSTM):



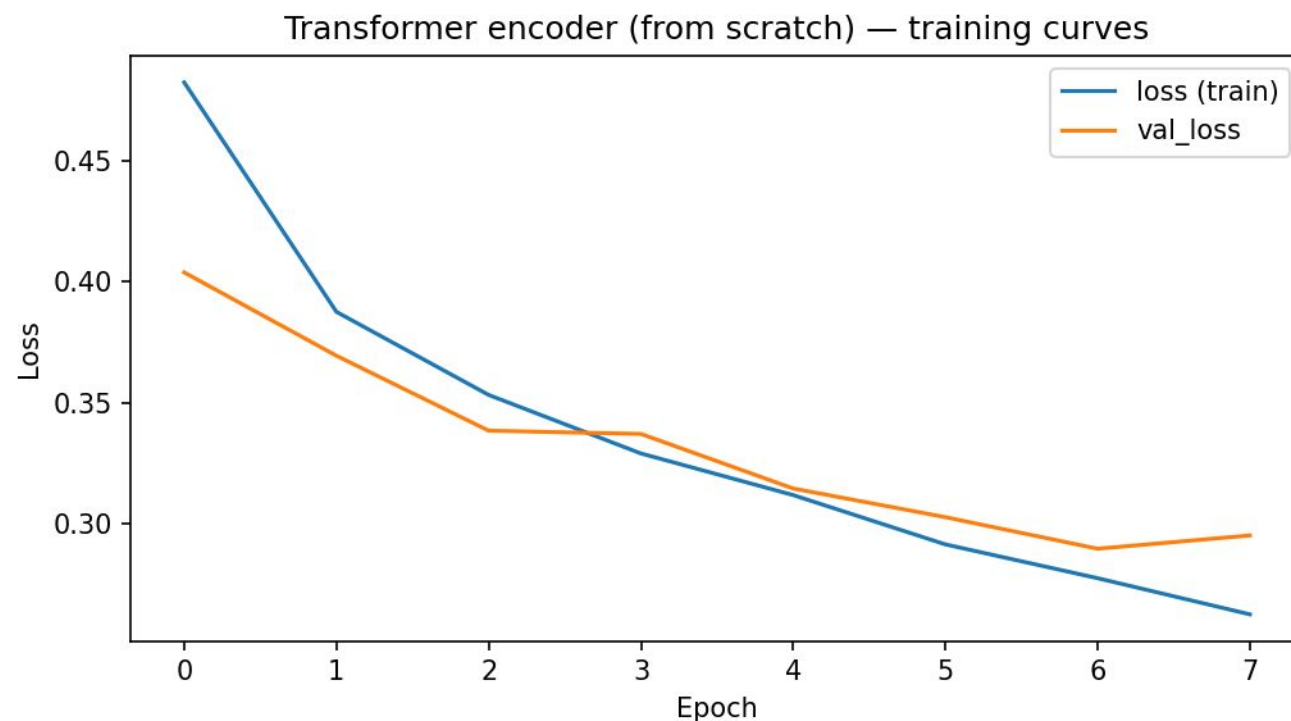
Classification report

	precision	recall	f1-score	support
fake	0.88	0.88	0.88	9091
real	0.88	0.88	0.88	9208
accuracy			0.88	18299
macro avg	0.88	0.88	0.88	18299
weighted avg	0.88	0.88	0.88	18299

- **Phase 2: Multi-Architecture Classification**

To rigorously distinguish real tweets from synthetic ones, we deployed a progression of five distinct architectures, ranging from traditional ML to state-of-the-art Deep Learning:

4. Transformer Encoder (From Scratch):



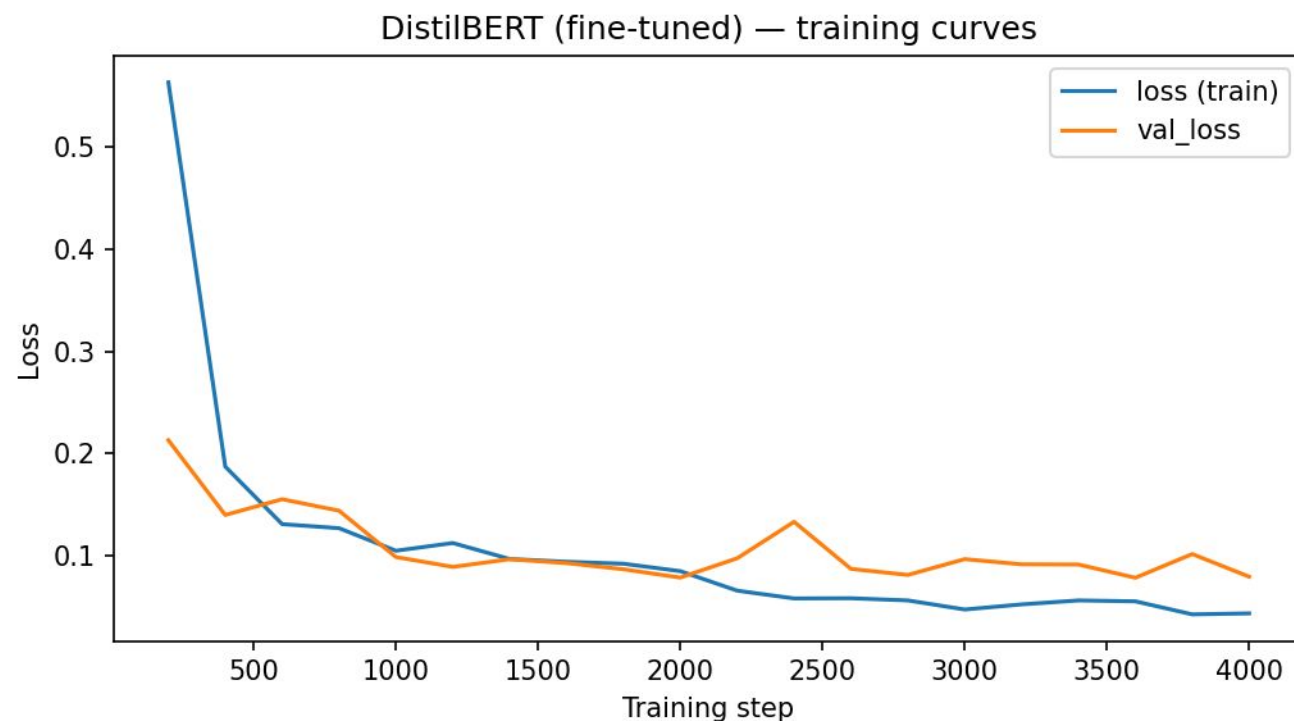
Classification report

	precision	recall	f1-score	support
fake	0.86	0.87	0.87	9091
real	0.87	0.86	0.87	9208
accuracy			0.87	18299
macro avg	0.87	0.87	0.87	18299
weighted avg	0.87	0.87	0.87	18299

- **Phase 2: Multi-Architecture Classification**

To rigorously distinguish real tweets from synthetic ones, we deployed a progression of five distinct architectures, ranging from traditional ML to state-of-the-art Deep Learning:

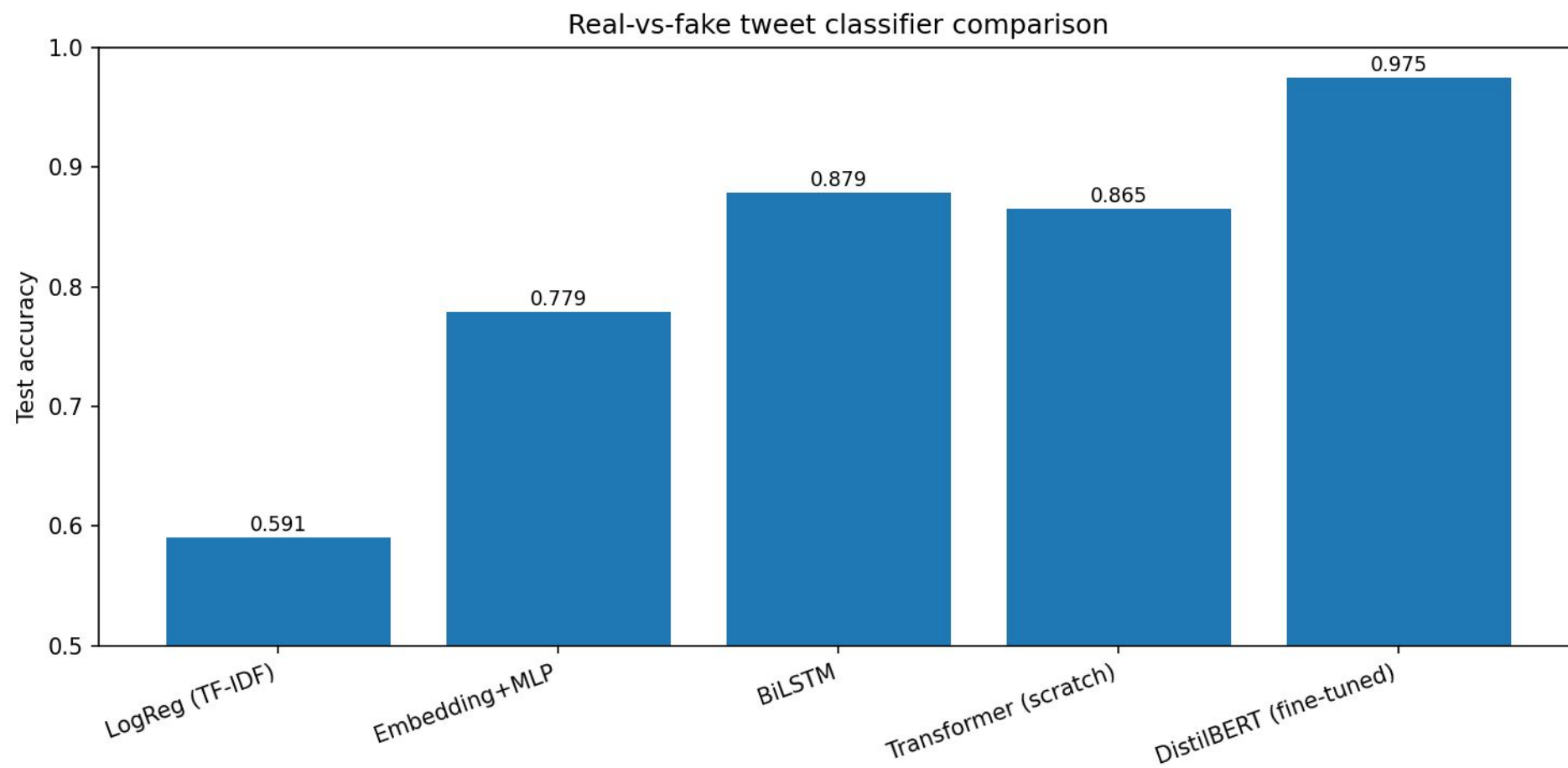
5. DistilBERT (Fine-Tuned):



Classification report				
	precision	recall	f1-score	support
fake	0.99	0.96	0.97	9091
real	0.96	0.99	0.98	9208
accuracy			0.97	18299
macro avg	0.98	0.97	0.97	18299
weighted avg	0.98	0.97	0.97	18299

- **Phase 2: Multi-Architecture Classification**

comparison_accuracy



- **Phase 3: Advanced LLM Synthesis**

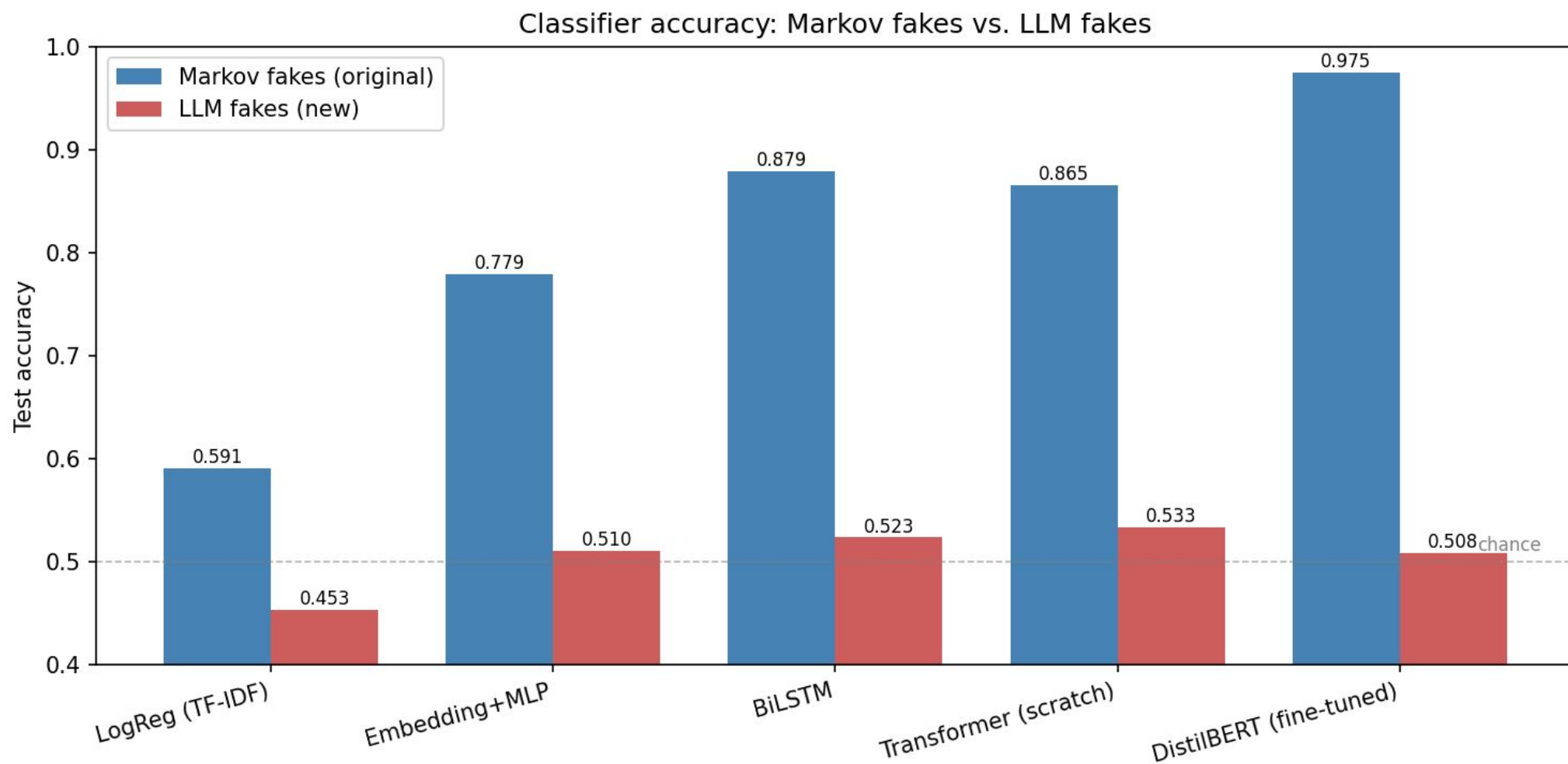
The question is:
do the patterns the classifiers learned generalize, or were they just learning to spot Markov-chain artifacts?

The Markov fakes are easy to spot. They're locally fluent but globally incoherent. A more realistic adversary is a small language model fine-tuned on the same tweets we're trying to imitate.

Smarter generator:

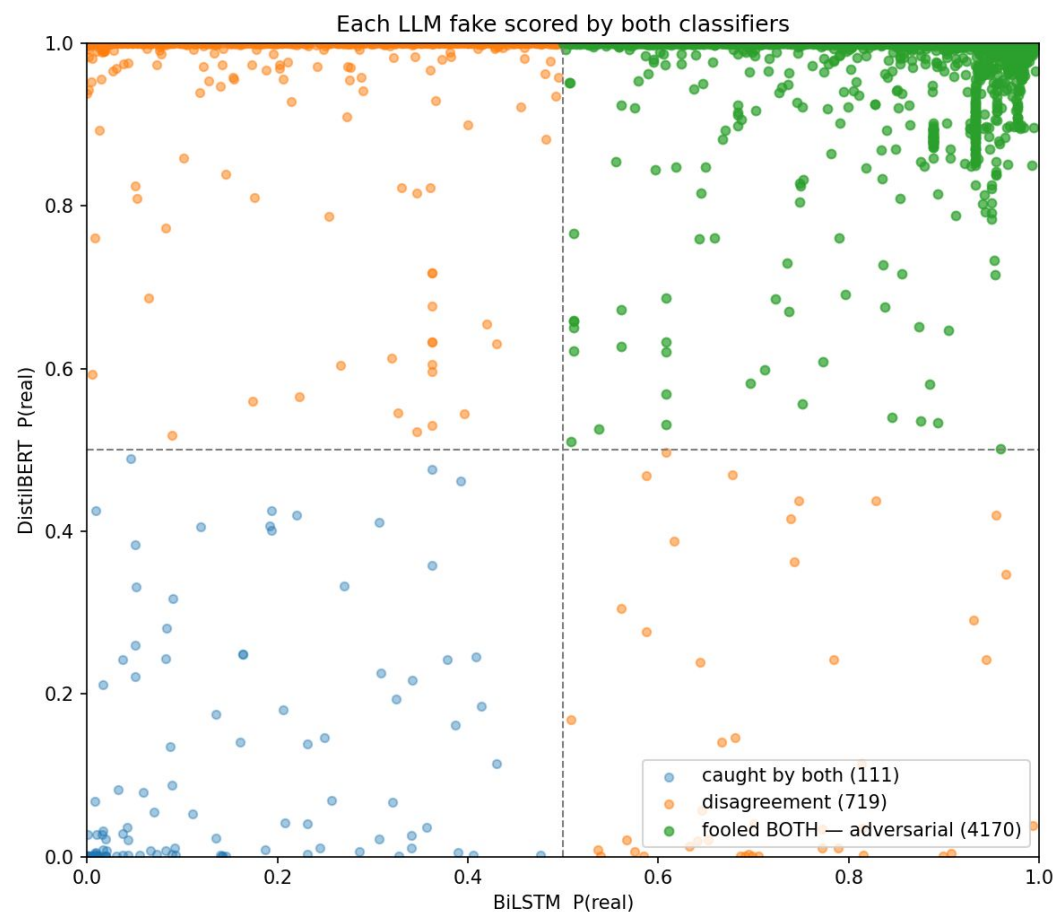
1. gpt2 (124M parameters)
2. Briefly fine-tune it on 10,000 real Trump tweets.
3. Sample 5,000 fake tweets from it with nucleus sampling

- **Phase 3: Advanced LLM Synthesis**



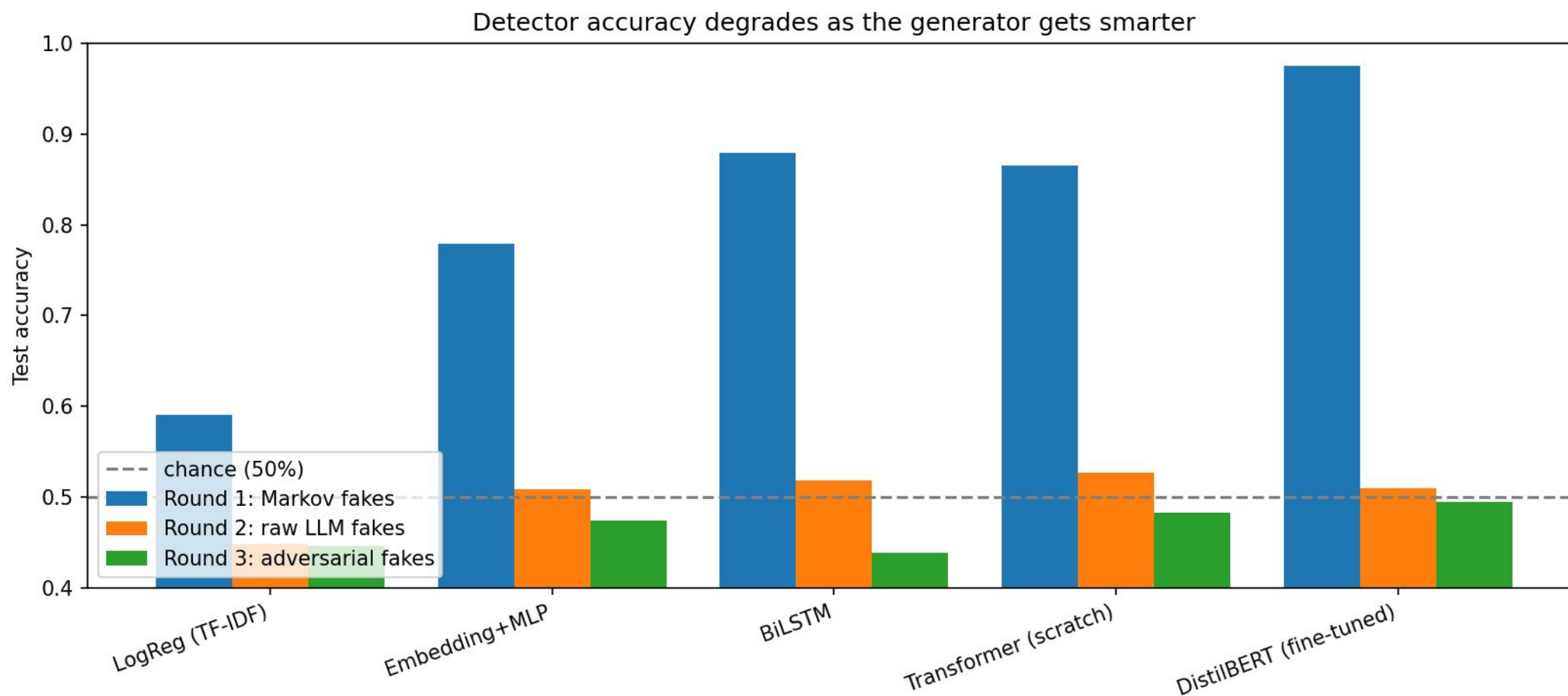
- **Phase 4: Ensemble Rejection Sampling and Stress Testing**

We took the output of our classifiers, feed it into the generator, and filter for even better tweets with our two strongest classifiers: BiLSTM and DistilBERT

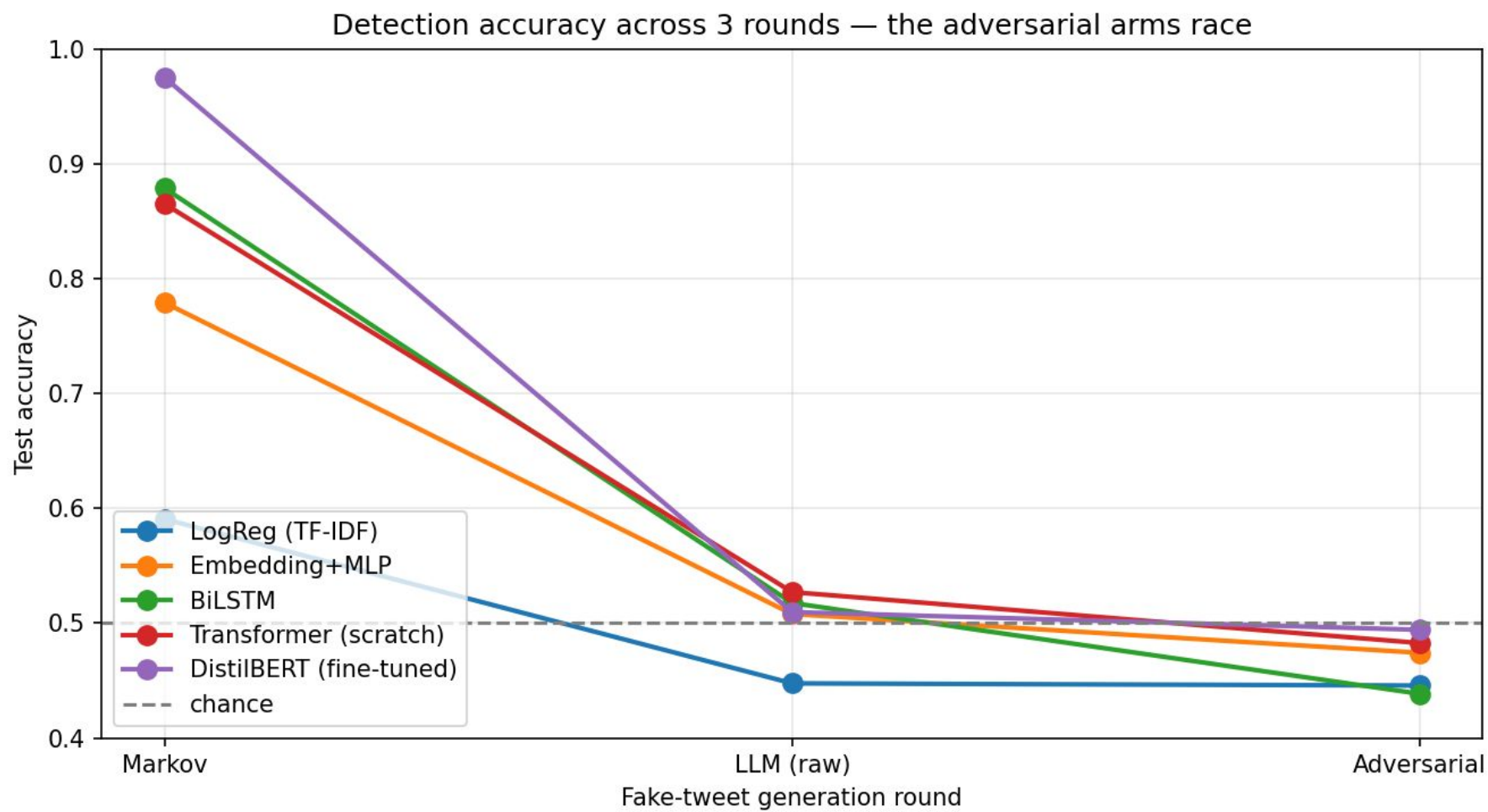


- **Phase 5: Re-classify Re-generated data**

In the previous work, we filtered the LLM fakes through a dual discriminator. Now we throw all 5 classifiers at this hard-mode set to see how each one degrades.



Discussion and Conclusion



Thanks for your attention